

SHANGYANG MIN

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EDUCATION

Brown University

Sep 2023 – May 2025

M.Sc. in Computer Science | GPA: 4.0/4.0 | Multimodal AI, Brain-Inspired Computing

Michigan State University

Aug 2019 – May 2023

B.S. in Computer Science Engineering, High Honor | GPA: 3.95/4.0 | ML, Computer Vision, Biomedical Imaging

EXPERIENCE

Hubei Lianhuashan AI Research Institute

Jun 2025 – Present

Research Engineer | Under Peking University Institute for General AI | Ezhou, China

- Architected an agent framework for traditional industries, enabling non-technical enterprises to deploy domain-specific agents through tool integration and deterministic workflow orchestration.
- Led executive workshops, technical bootcamps, and university collaborations with Hubei institutions across enterprise and academic audiences.

Abysen AI

Feb 2025 – Present

Founder | Agent Infrastructure | United States | [abysen.com](#)

- Founded an agent infrastructure startup developing general-purpose agent orchestration frameworks for both research and enterprise use cases.

Lee Lab, Brown University

Sep 2024 – May 2025

Research Assistant | Brain-Computer Interface & Multimodal Interaction

- Built a real-time EEG-to-Unity pipeline for brain-signal avatar control and developed a vision + EEG fusion architecture for cross-modal interaction.

HAAI Lab, Michigan State University

May 2022 – Aug 2023

Research Assistant | Biomedical Imaging | Funded by NSF & HFH+MSU Cancer Grant

- Proposed Feature Imitating Networks (FINs) that distill large-model representations into lightweight CNNs, cutting inference time by 3× while improving accuracy—published at IEEE EMBC 2024 with open-source release.

Henry Ford Health × Michigan State University

Sep 2022 – Sep 2023

Research Assistant | NIH-Funded Program | Tumor Detection & Radiomics

- Built ML pipelines for DCE-MRI tumor detection and radiomics feature extraction; standardized preprocessing for multi-institutional model evaluation.

SELECTED PROJECTS

Tethro: Realtime Voice Agent and Relationship OS

- End-to-end real-time voice agent based on achievements from Abysen AI—handling relationship memory, tool calls, and long-context conversation state.
- Designed a lower-latency streaming voice architecture using partial-ASR READ prefetch, final-gated answer streaming, barge-in cancellation, and watchdog/owner gates, with full-flow covering READ, WRITE, social, and interruption paths.

LLM-Agent-Core: Open-Source LLM & Agent Curriculum

- Designed and maintained a hands-on AI course with runnable notebooks spanning ML, Transformers, SFT/DPO, and agentic systems; implemented Agent workflows through a unified LLM backend abstraction.

Trigger-Simulation Backdoor Removal Defense (TSBRD)

- Proposed virtual-prompt realignment for backdoored LLMs, reducing attack success rate from 93% → 0% on Llama-2 within 7 epochs while preserving MMLU accuracy.

VQA with Action Insights

- Integrated SOTA vision models as tokenized video-info annotations into LLM prompts, achieving fine-tuned SeViLa-level performance with zero-shot GPT-4 inference.

scCST: Continuous Spatial-Temporal Transformer

- Developed a transformer for interpolating cellular trajectories in single-cell sequencing data, enabling continuous modeling of gene expression dynamics.

Additional research projects: shangyangmin.com/projects

TECHNICAL SKILLS

Core: Python, PyTorch, TypeScript, Hugging Face (Transformers/PEFT/TRL), C/C++, SQL

Systems: FastAPI, Docker, Redis, PostgreSQL/pgvector, vLLM, Ollama, CUDA, Git, Linux, W&B, LaTeX

Focus: LLM/Agent Systems, Realtime Voice AI, Multimodal Learning, Computer Vision, Diffusion Models, BCI

PUBLICATION

S. Min, H. B. Ebadian, T. Alhanai, M. M. Ghassemi, “Feature Imitating Networks Enhance the Performance, Reliability and Speed of Deep Learning on Biomedical Image Processing Tasks,” IEEE EMBC 2024.